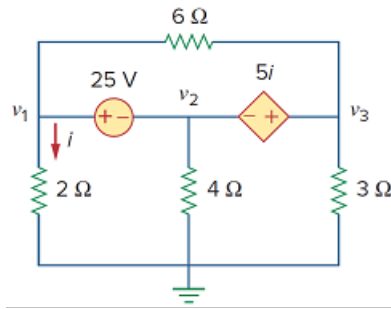


Problem

Find v_1 , v_2 , and v_3 in the circuit of Fig. 3.14 using nodal analysis.

Figure 3.14:



Step-by-step solution

Step 1 of 5

Refer to Figure 3.14 from the textbook for circuit diagram.

Observe that nodes v_1 and v_2 form a super node and nodes v_2 and v_3 form another super node. Nodes v_1 , v_2 , and v_3 form an overall super node.

The modified circuit diagram with current directions is shown in Figure 1.

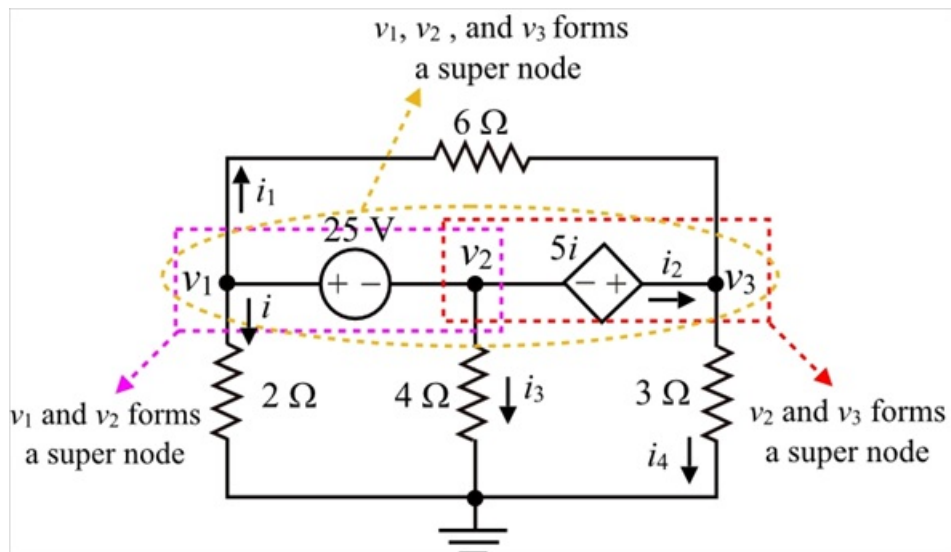


Figure 1

Step 2 of 5

From the Figure 1, observe that,

$$v_1 - v_2 = 25$$

$$v_1 = 25 + v_2 \dots\dots (1)$$

And,

$$v_3 - v_2 = 5i \dots\dots (2)$$

Observe that nodes v_1, v_2 , and v_3 form an overall super node. Apply Kirchhoff's current law to overall super node.

$$i + i_1 + i_3 + i_4 - i_1 = 0$$

Express currents in node voltages.

$$\frac{v_1}{2} + \frac{v_1 - v_3}{6} + \frac{v_2}{4} + \frac{v_3}{3} + \frac{v_3 - v_1}{6} = 0$$

$$\frac{v_1}{2} + \frac{v_2}{4} + \frac{v_3}{3} = 0$$

$$\frac{6v_1 + 3v_2 + 4v_3}{12} = 0$$

$$6v_1 + 3v_2 + 4v_3 = 0 \dots\dots (3)$$

Step 3 of 5

Substitute equation (1) in equation (3).

$$6(25 + v_2) + 3v_2 + 4v_3 = 0$$

$$150 + 6v_2 + 3v_2 + 4v_3 = 0$$

$$9v_2 + 4v_3 = -150$$

$$9v_2 = -150 - 4v_3$$

$$v_2 = -\left(\frac{150 + 4v_3}{9}\right) \dots\dots (4)$$

Recall the equation (2).

$$v_3 - v_2 = 5i$$

But the current i from the Figure 1 is,

$$i = \frac{v_1}{2}$$

Substitute $\frac{v_1}{2}$ for i in equation (2).

$$v_3 - v_2 = 5\left(\frac{v_1}{2}\right) \left(\text{since } v_1 = 25 + v_2\right)$$

$$v_3 - v_2 = 5\left(\frac{25 + v_2}{2}\right)$$

$$2v_3 - 2v_2 = 125 + 5v_2$$

$$7v_2 - 2v_3 = -125 \dots\dots (5)$$

Substitute equation (4) in equation (5).

$$7\left[-\left(\frac{150 + 4v_3}{9}\right)\right] - 2v_3 = -125$$

$$-1050 - 28v_3 - 18v_3 = -1125$$

$$46v_3 = 75$$

$$v_3 = 1.63 \text{ V}$$

Therefore, the node voltage v_3 is 1.63 V.

Step 4 of 5

Substitute 1.63 V for v_3 in equation (4).

$$\begin{aligned} v_2 &= -\left(\frac{150+4(1.63)}{9}\right) \\ &= -\frac{156.52}{9} \\ &= -17.39 \text{ V} \end{aligned}$$

Therefore, the node voltage v_2 is -17.39 V.

Step 5 of 5

Substitute -17.39 V for v_2 in equation (1).

$$\begin{aligned} v_1 &= 25 - 17.39 \\ &= 7.61 \text{ V} \end{aligned}$$

Therefore, the node voltage v_1 is 7.61 V.